

Hepker, Sprout - Final Project

05/04/2025

1) Define The Problem:

This project focuses on forecasting short-term player engagement on Steam using a curated sample of popular games. Accurate daily player count forecasts can help inform real-world decisions like server load management, maintenance windows, and when to time promotions or releases. The goal is to model patterns in player activity that repeat weekly or shift during unique periods, like the early COVID-19 lockdown.

Two approaches are used to forecast the top-level (aggregated) series: a manual SARIMA model that includes time-based dummy variables (weekends and lockdown), and a hierarchical time series (HTS) method that takes advantage of the genre and game-level structure underneath the total.

2) Gathering Information

The data for this project comes from a publicly available Steam dataset on Mendeley Data, which includes detailed player count and price history for around 2,000 games between late 2017 and mid-2020. For this project, I narrowed it down to six games across three genres (Indie, RPG, MMO). Daily player counts were created from 5-minute and hourly logs that were combined and averaged to produce daily time series for each title. Data on game names, genres, and app IDs was merged in and two dummy variables to capture relevant patterns in the data were created: one for weekends and one for the early COVID-19 lockdown period (March 15 – May 15, 2020). The final dataset covers a 15-month span from May 2019 to August 2020 and was structured specifically to support both traditional and hierarchical forecasting methods.

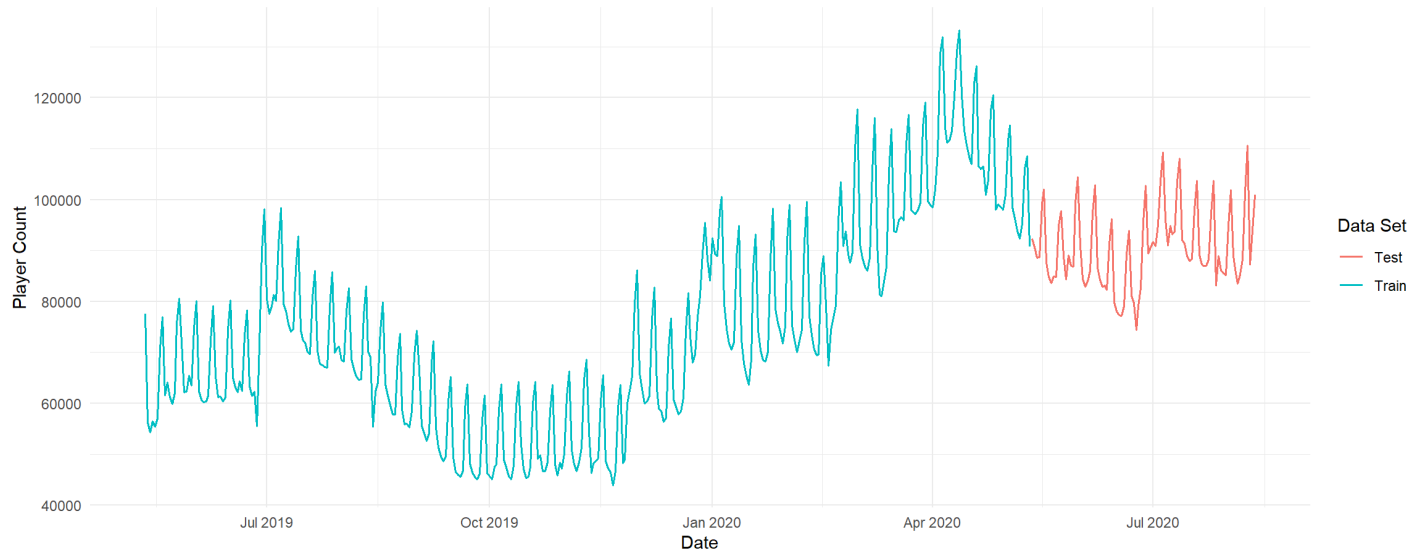
3) Explore and Visualize Series

To better understand the structure of the data, I began by plotting the overall trend in daily player counts across all games, with clear delineation between the training and test periods. This was followed by separate visualizations at the genre and individual game levels to highlight recurring patterns such as weekly seasonality and shifts in engagement. Basic summary statistics were also calculated for the top-level series to provide a snapshot of its scale and variability. Notably, player counts appear to rise during the early COVID-19 lockdown period, suggesting a temporary change in user behavior.

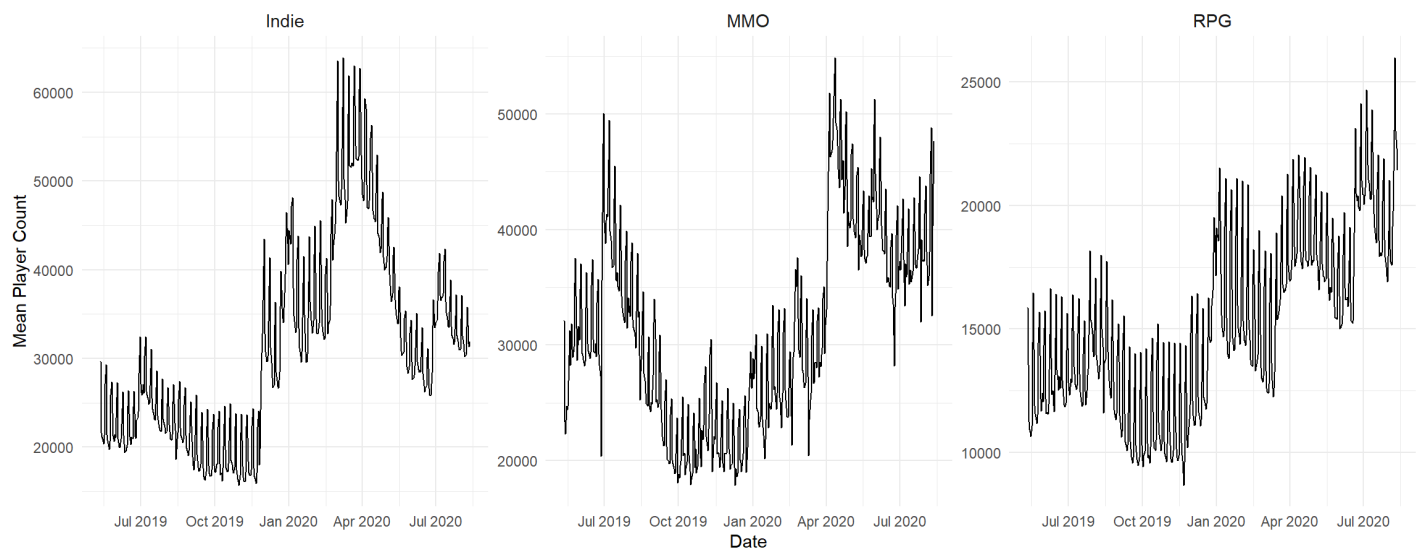
Summary Statistics for Total Daily Player Count

Statistic	Value
Mean	77668.35
SD	19572.81
Min	43970.16
Max	133267.37

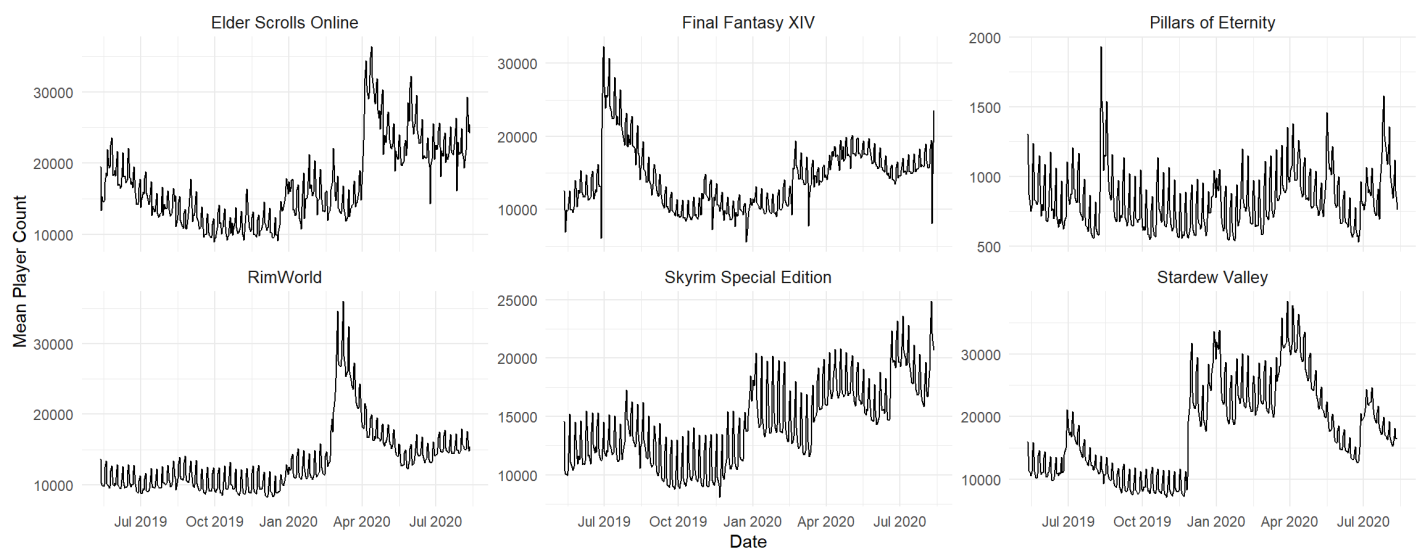
Train and Test Series – Total Player Count



Daily Mean Player Count by Genre



Daily Mean Player Count by Game



4) Choosing and Fitting Models

Benchmark Methods

Among benchmark models, the Naïve forecast produced the lowest RMSE, MAE, and MAPE values. Therefore, it is used as the baseline for comparison against more complex models.

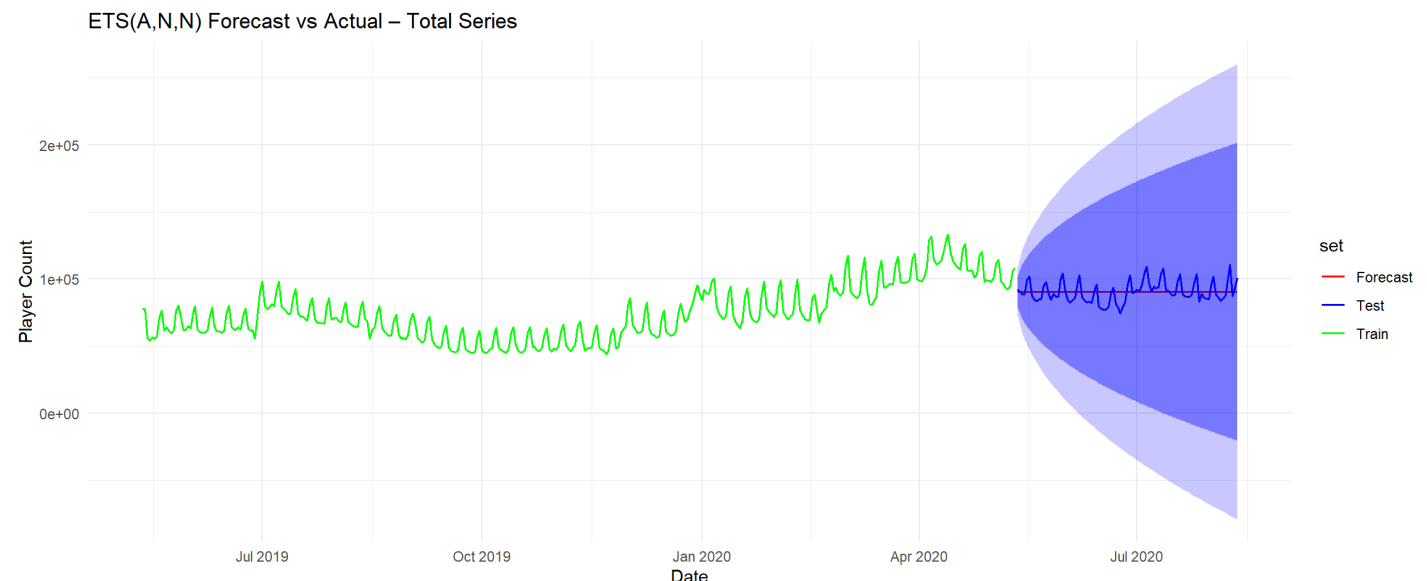
Benchmark Forecast Accuracy (Top-Level Series)

Method	RMSE	MAE	MAPE
Naive	7787.137	6295.76	6.937
Seasonal Naive	21964.999	20052.03	21.959
Mean	17897.078	16115.49	17.210
Drift	7901.686	6636.94	7.442

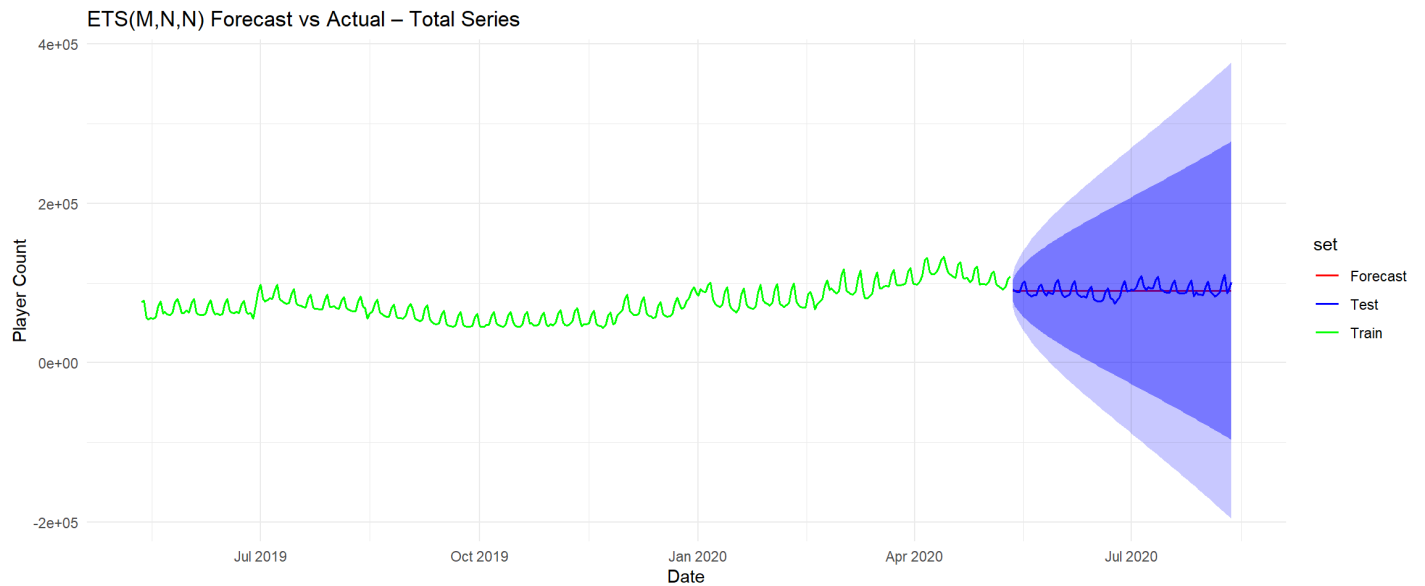
ETS Models

Among the ETS models evaluated, ETS(M,N,N) showed the lowest RMSE and MAPE values and was selected by the auto-ETS procedure. This model assumes a multiplicative error and no trend or seasonal component. Other ETS variants with trend components (A,A,N), (M,A,N), and (M,M,N) performed significantly worse, suggesting that trend modeling introduced instability into the forecasts. ETS(M,N,N) marginally outperformed the naïve benchmark, indicating that exponential smoothing may provide a slight measurable improvement in forecast accuracy.

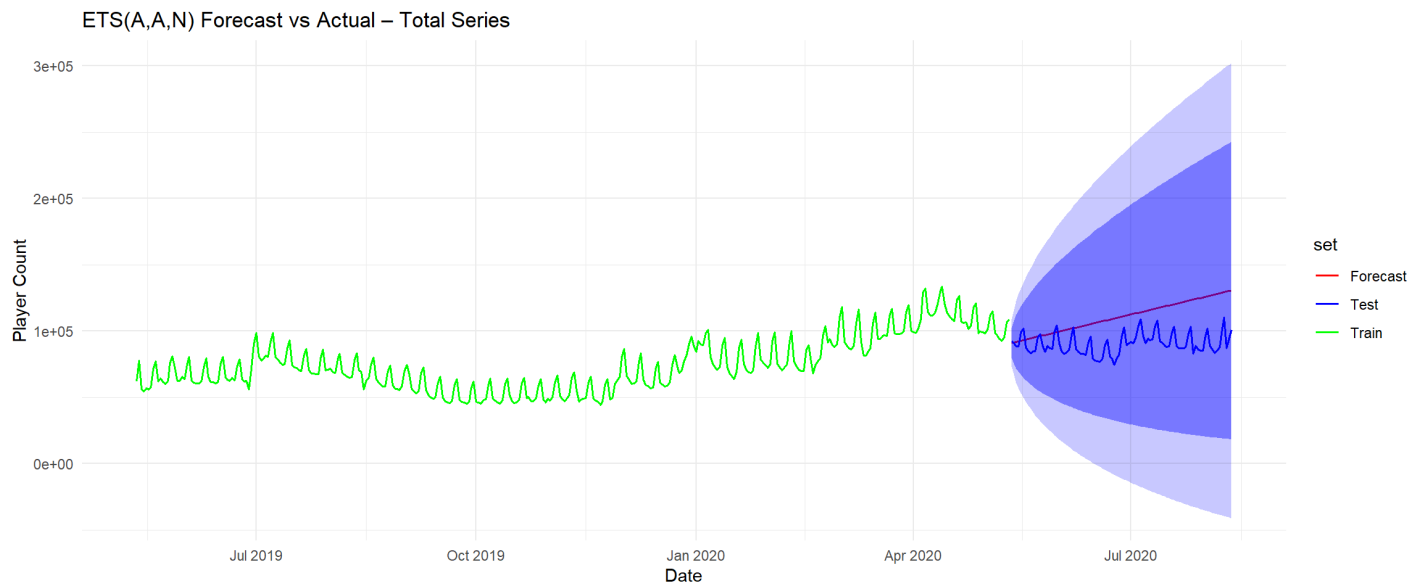
ETS Forecast Plot – ETS(A,N,N)



ETS Forecast Plot – ETS(M,N,N)



ETS Forecast Plot – ETS(A,A,N)



ETS Model Accuracy Comparison

ETS Model Accuracy Comparison – Top-Level Series

Model	RMSE	MAE	MAPE	AICc
ETS(A,N,N)	7787.185	6296.048	6.938	8826.404
ETS(M,N,N)	7787.185	6296.050	6.938	8782.664
ETS(A,A,N)	23993.562	20873.563	23.697	8834.207

ETS Estimation Equations

ETS(A,N,N) — Additive Error, No Trend, No Seasonality

$$y_t = \ell_{t-1} + \varepsilon_t \quad (\text{Observation})$$

$$\ell_t = \ell_{t-1} + \alpha \varepsilon_t \quad (\text{Level})$$

$$\hat{y}_{t+h|t} = \ell_t \quad (\text{Forecast})$$

ETS(M,N,N)/(auto) — Multiplicative Error, No Trend, No Seasonality

$$y_t = \ell_{t-1}(1 + \varepsilon_t) \quad (\text{Observation})$$

$$\ell_t = \ell_{t-1}(1 + \alpha \varepsilon_t) \quad (\text{Level})$$

$$\hat{y}_{t+h|t} = \ell_t \quad (\text{Forecast})$$

ETS(A,A,N) — Additive Error, Additive Trend, No Seasonality

$$y_t = \ell_{t-1} + b_{t-1} + \varepsilon_t \quad (\text{Observation})$$

$$\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t \quad (\text{Level})$$

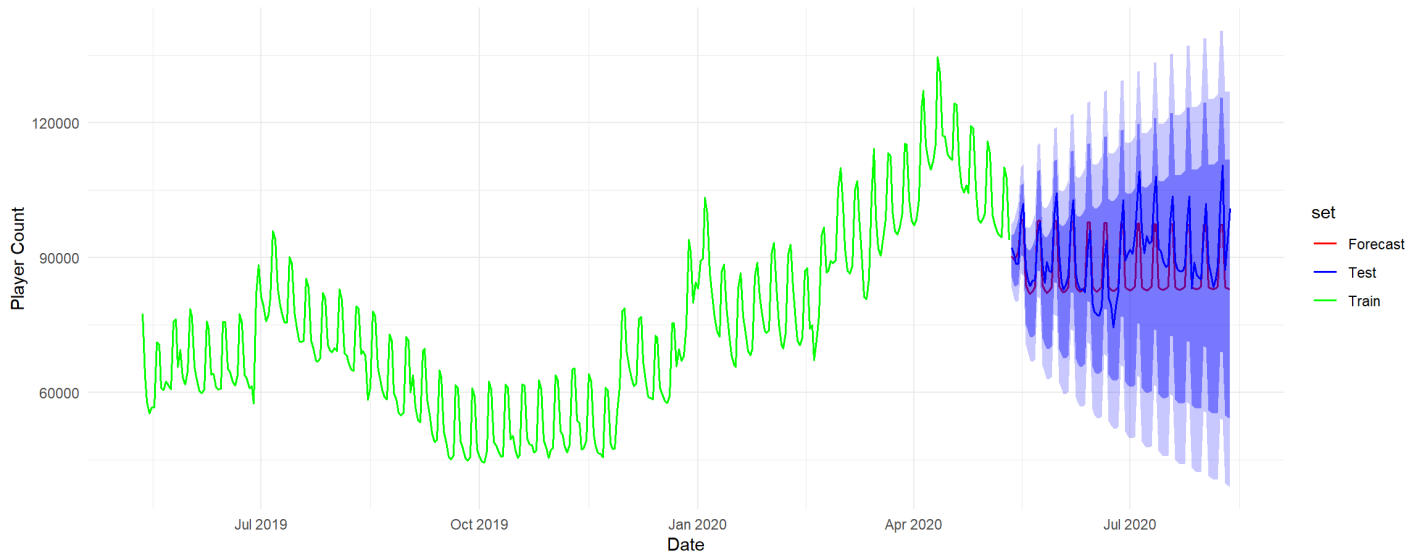
$$b_t = b_{t-1} + \beta \varepsilon_t \quad (\text{Trend})$$

$$\hat{y}_{t+h|t} = \ell_t + hb_t \quad (\text{Forecast})$$

ARIMA Models

Auto Arima Benchmark

Auto ARIMA Forecast vs Actual – Total Series (with Dummies)

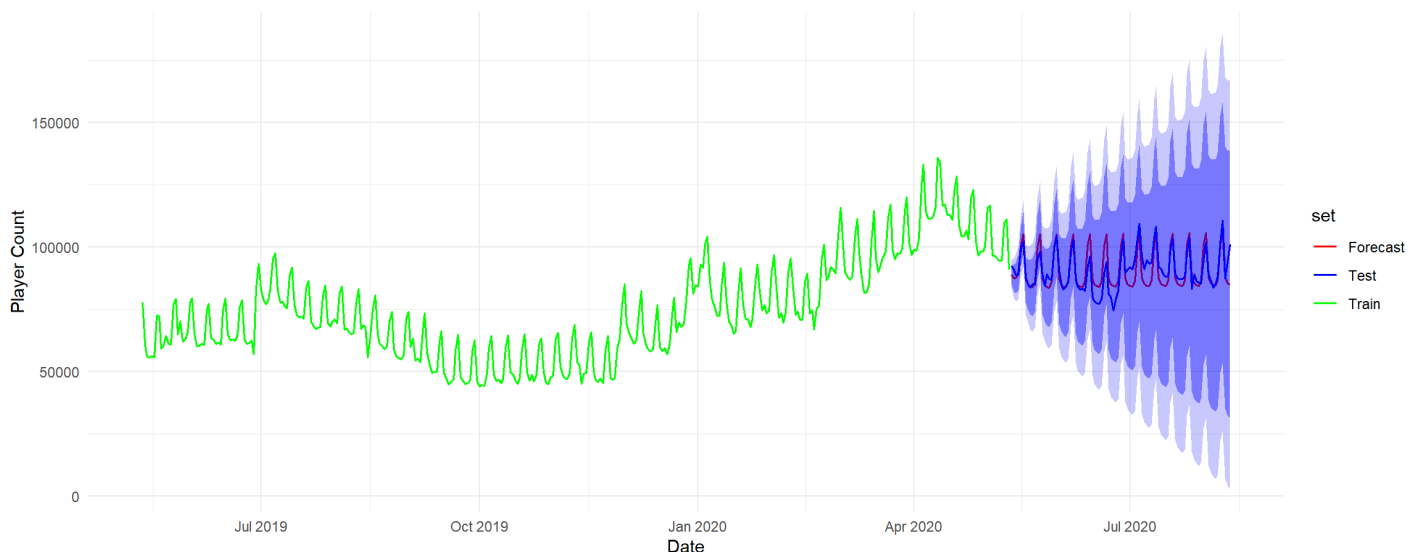


Manual ARIMA Selection

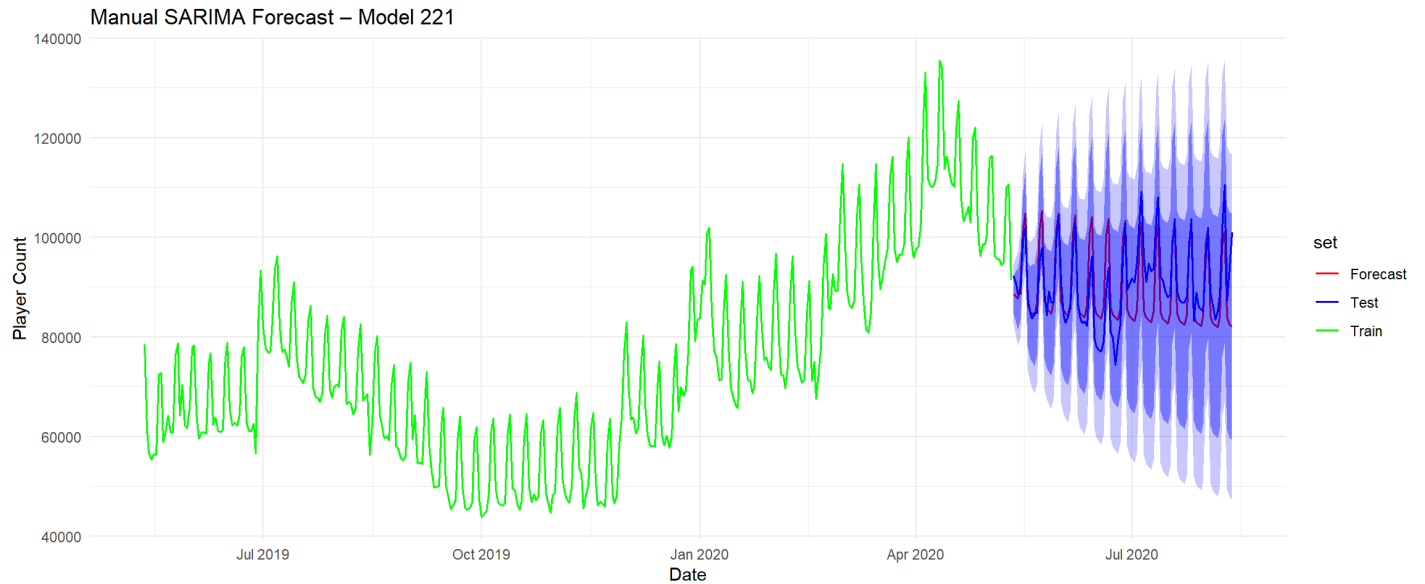
To identify suitable ARIMA models, I conducted a comprehensive grid search across $SARIMA(p,d,q)(P,D,Q)[7]$ configurations using external regressors for weekends and the COVID lockdown period. For each model, I recorded forecast accuracy (RMSE, MAPE, AICc) as well as residual autocorrelation using the Ljung-Box test at lag 14. Models were ranked by both RMSE and p-value to assess the tradeoff between predictive accuracy and model diagnostics. Model 218 ($SARIMA(3,1,3)(2,0,1)[7]$) offered the lowest RMSE overall. Model 221 ($SARIMA(2,0,0)(3,0,1)[7]$) was not the top in terms of residual diagnostics but stood out in the top p-value set due to its significantly better forecast accuracy (lowest RMSE, MAE, and MAPE in that group). These two models were selected for their complementary strengths, model 218 for accuracy and model 221 for balancing strong accuracy with low residual autocorrelation.

Manual SARIMA (Minimized RMSE)

Manual SARIMA Forecast – Model 218



Manual SARIMA (Minimized Autocorrelation)



ARIMA Model Accuracy Comparison

ARIMA Model Accuracy Comparison – Top-Level Series

Model	RMSE	MAPE	AICc	Ljung_Box_p
Auto ARIMA	5836.960	5.092	7027.289	0.000
SARIMA(3,1,3)(2,0,1)[7]	4981.576	4.245	6904.127	0.062
SARIMA(2,0,0)(3,0,1)[7]	5555.702	4.866	6927.965	0.235

ARIMA and SARIMA Model Equations

Auto ARIMA – ARIMA(2,1,3)

This model includes 2 non-seasonal AR terms, 1 non-seasonal difference, and 3 non-seasonal MA terms.

$$(1 - \phi_1 B - \phi_2 B^2)(1 - B)y_t = (1 + \theta_1 B + \theta_2 B^2 + \theta_3 B^3)\varepsilon_t$$

Model 218 – ARIMA(3,1,3)(2,0,1)[7]

This model includes 3 non-seasonal AR terms, 1 non-seasonal difference, 3 non-seasonal MA terms, 2 seasonal AR terms, and 1 seasonal MA term with period 7.

$$(1 - \phi_1 B - \phi_2 B^2 - \phi_3 B^3)(1 - B)(1 - \Phi_1 B^7 - \Phi_2 B^{14})y_t \\ = (1 + \theta_1 B + \theta_2 B^2 + \theta_3 B^3)(1 + \Theta_1 B^7)\varepsilon_t$$

Model 221 – ARIMA(2,0,0)(3,0,1)[7]

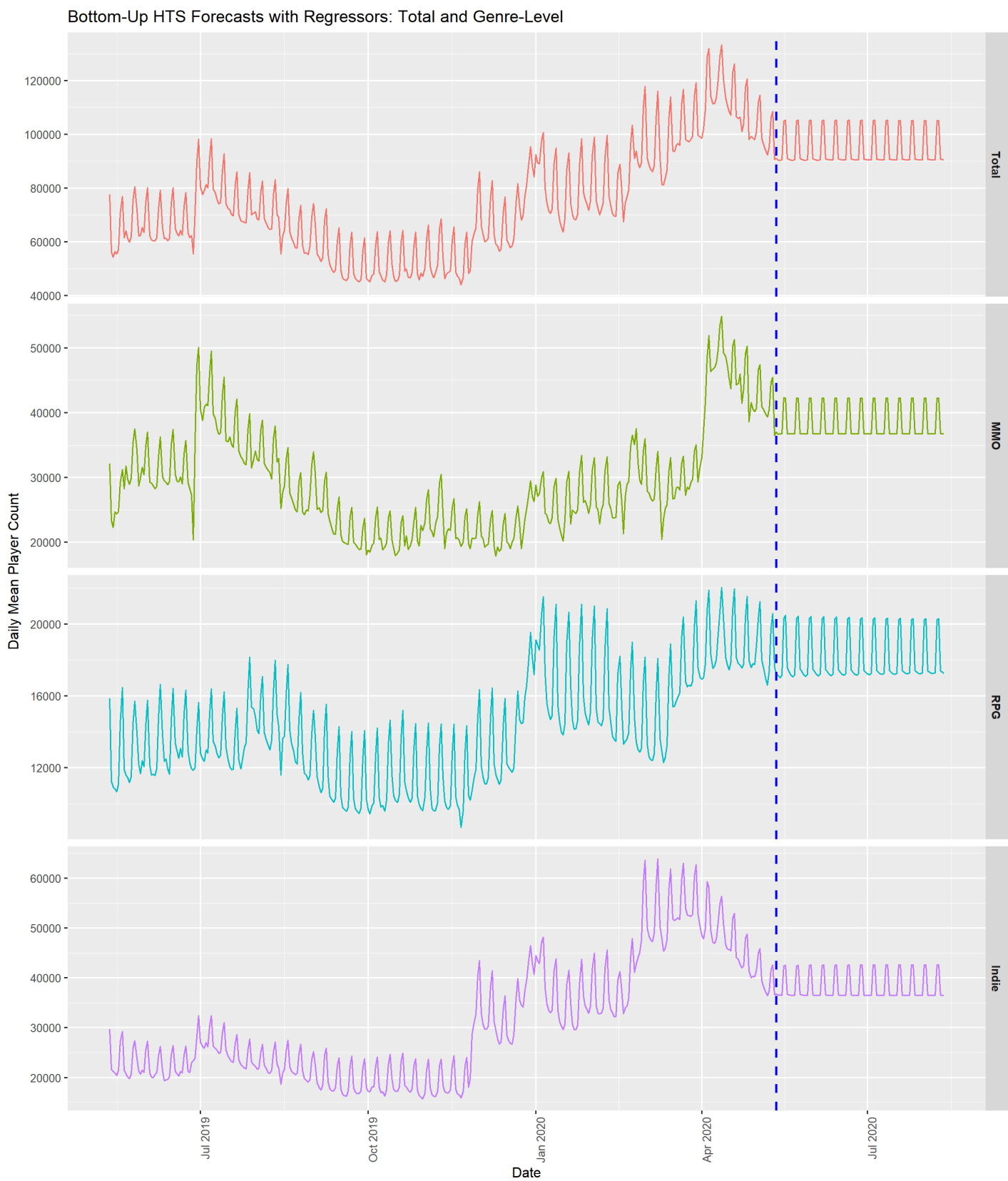
This model includes 2 non-seasonal AR terms, 3 seasonal AR terms, and 1 seasonal MA term, with no differencing.

$$(1 - \phi_1 B - \phi_2 B^2)(1 - \Phi_1 B^7 - \Phi_2 B^{14} - \Phi_3 B^{21})y_t = (1 + \Theta_1 B^7)\varepsilon_t$$

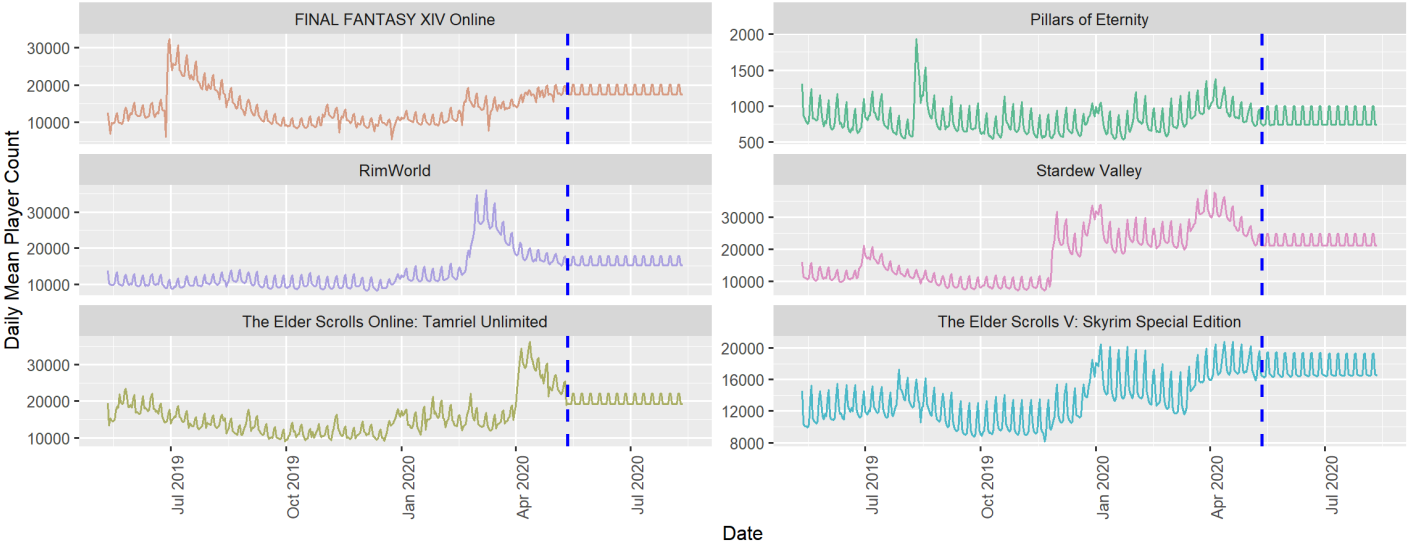
HTS Models

A hierarchical time series (HTS) structure was applied to capture forecasts at both the game-level and genre-level, including the overall total. Two reconciliation approaches were tested. The Bottom-Up method produced forecasts for each individual game and then aggregated them to higher levels, operating under the assumption that the most detailed data contains the most reliable signal. In contrast, the MinT (Minimum Trace) method combined base forecasts from all levels using a generalized least squares framework, minimizing the total forecast error variance while ensuring aggregate consistency.

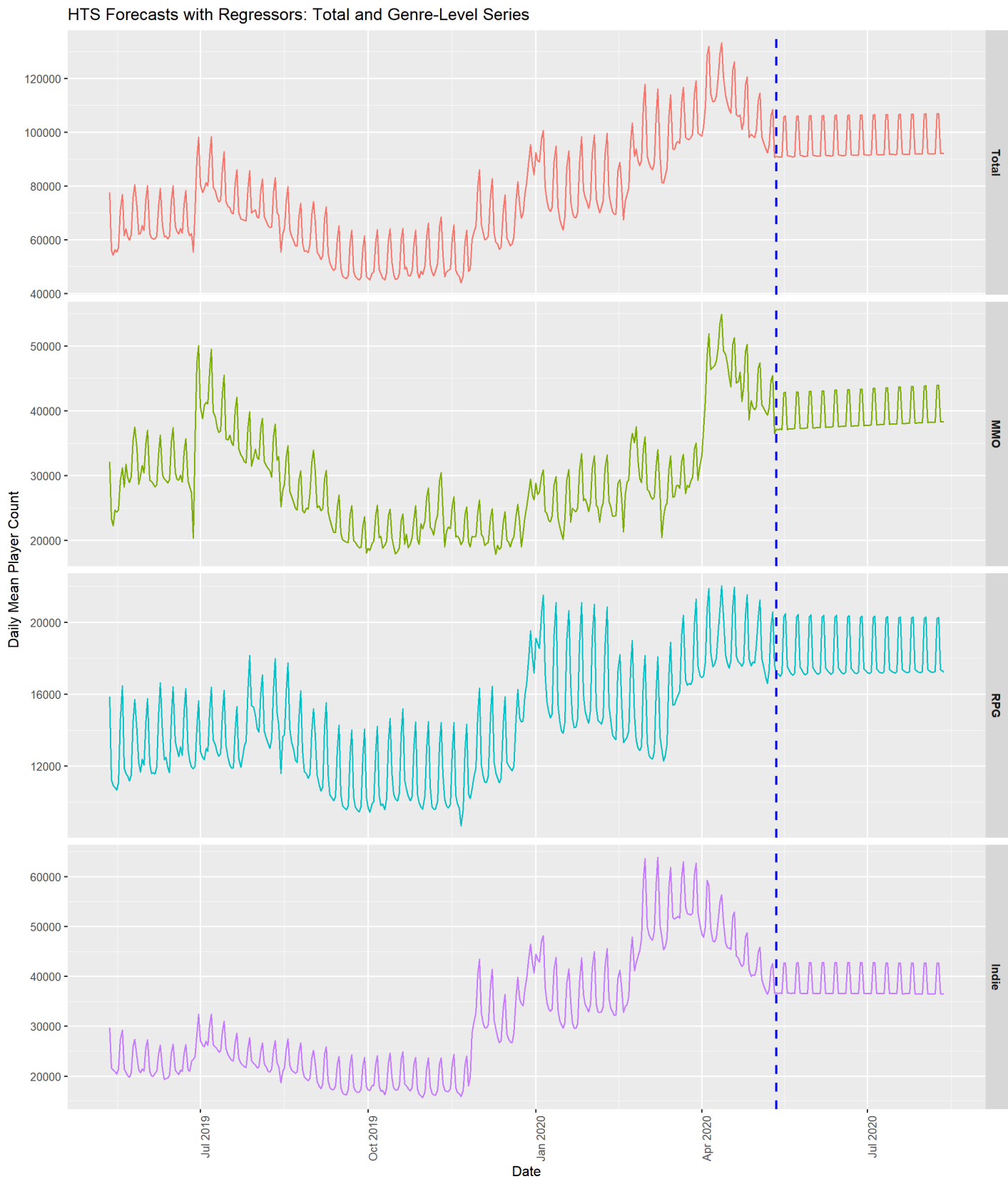
Bottom Up Method



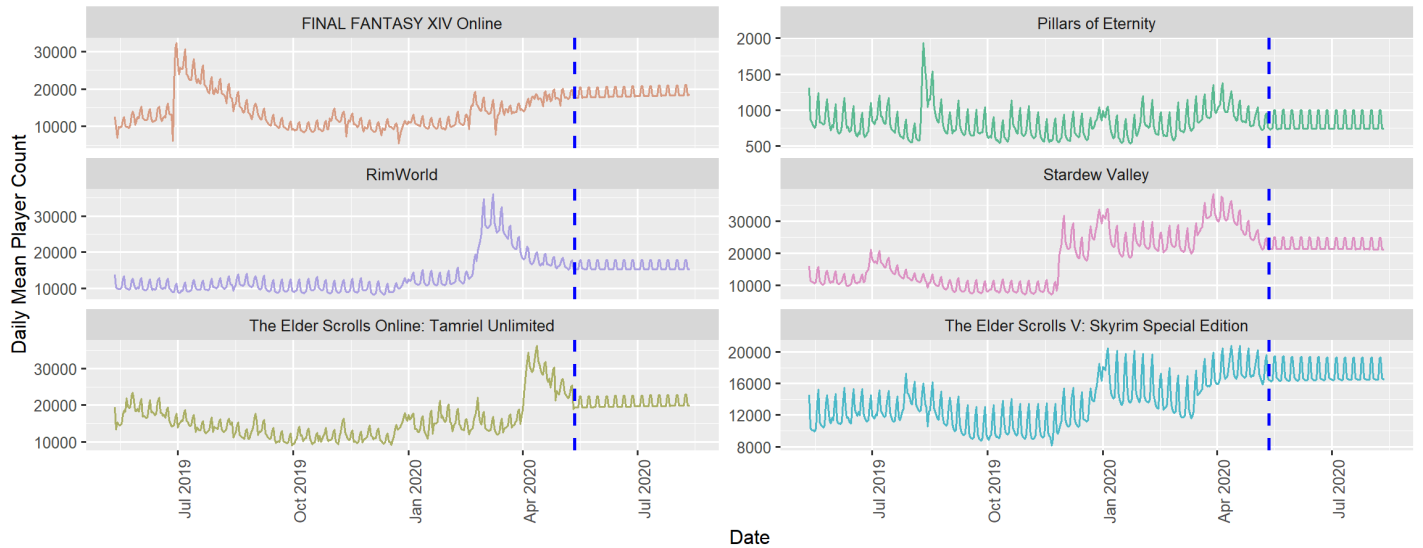
Bottom-Up HTS Forecasts: Game-Level (Level 2)



MinT Method



HTS Forecasts: Game-Level (Level 2) Series



Hierarchical Forecasting Equations

Bottom-Up (BU) Forecasting

The **bottom-up method** generates coherent forecasts by first forecasting only the bottom-level series and then aggregating these to obtain forecasts for all higher levels in the hierarchy.

The bottom-up reconciled forecasts are given by:

$$\tilde{\mathbf{y}}_h = \mathbf{S}\hat{\mathbf{b}}_h$$

MinT (Minimum Trace) Reconciliation

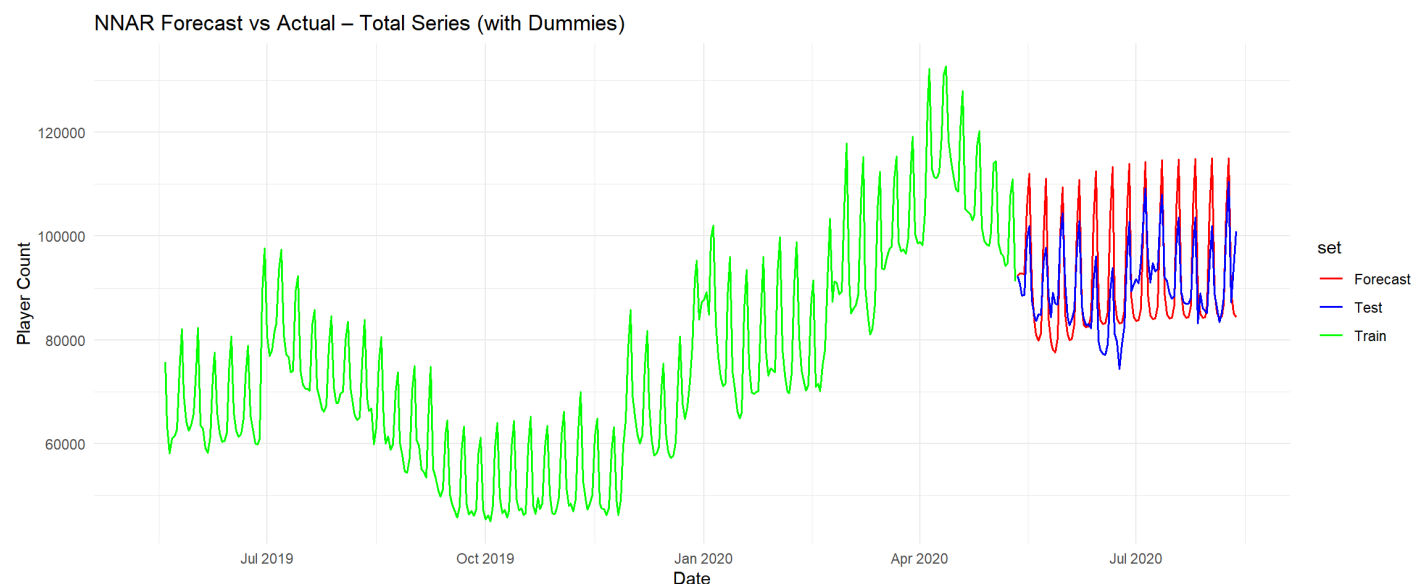
The MinT method reconciles base forecasts from all series (at all levels) to ensure coherence. It minimizes the total forecast error variance across the hierarchy using a generalized least squares approach.

The MinT reconciled forecasts are given by:

$$\hat{\mathbf{y}}_{t+h}^* = \mathbf{S}(\mathbf{S}'\mathbf{W}_h^{-1}\mathbf{S})^{-1}\mathbf{S}'\mathbf{W}_h^{-1}\hat{\mathbf{y}}_{t+h}$$

NNAR Model

After manually experimenting with different parameters, a NNAR model was fit using weekly lags ($p = 7$) and 10 hidden nodes (size = 10). Two binary regressors (weekend and COVID dummies) were included to capture calendar effects. These adjustments improved alignment with seasonal patterns and reduced overestimation.



Neural Network Autoregression (NNAR) Equations

The NNAR model is a type of feed-forward neural network applied to lagged values of the time series. In this case, the model used 7 lagged observations as inputs and 10 hidden nodes, along with binary dummy regressors.

Each hidden node takes a weighted combination of the inputs:

$$z_j = b_j + \sum_{i=1}^{p+k} w_{i,j} x_i$$

This is passed through a sigmoid activation function to produce the node output. The final forecast is then a weighted sum of all hidden node outputs.

Forecast Comparison

Forecast Accuracy – Top-Level Series (ARIMA, ETS, Reconciliation, and NNAR Methods)

Method	RMSE	MAE	MAPE
Naive Benchmark	7787.137	6295.760	6.937
ARIMA(2,3,0) (Auto)	5836.960	4687.425	5.092
ARIMA(3,1,3)(2,0,1)[7]	4981.576	3831.618	4.245
ARIMA(2,0,0)(3,0,1)[7]	5555.702	4407.365	4.866
ETS(A,N,N)	7787.185	6296.048	6.938
ETS(M,N,N)	7787.185	6296.050	6.938
ETS(A,A,N)	23993.562	20873.563	23.697
Bottom-Up	6473.657	5234.576	5.981

Method	RMSE	MAE	MAPE
MinT	7154.238	5915.117	6.758
NNAR(p=7, size=10)	6842.957	5481.538	5.971

Step 4b) SARIMA Model Tweaks

Forecast Accuracy Summary

Forecast Accuracy – ARIMA Model Tweaks (Including Baseline)

Model	Description	RMSE	MAE	MAPE	AICc	Ljung_Box_p
Baseline	ARIMA(3,1,3)(2,0,1) [7]	4981.576	3831.618	4.245	6904.127	0.062
Simplified Variant	ARIMA(2,1,2)(2,0,1) [7]	5203.190	4021.316	4.443	6913.571	0.037
Reduce MA Order	ARIMA(3,1,2)(2,0,1) [7]	5193.396	4010.090	4.429	6901.990	0.136
Reduce AR Order	ARIMA(2,1,3)(2,0,1) [7]	5005.467	3860.079	4.277	6901.941	0.114
Reduce Seasonal AR	ARIMA(3,1,3)(1,0,1) [7]	5530.331	4302.764	4.743	6913.104	0.006
Reduce Seasonal MA	ARIMA(3,1,3)(2,0,0) [7]	5150.913	3985.852	4.513	6969.778	0.000
No Regressors	ARIMA(3,1,3)(0,0,0) [7]	12834.956	11202.690	13.010	7484.840	0.000
Weekend Dummy Only	ARIMA(3,1,3)(2,0,1) [7]	6619.425	5448.893	6.141	6903.476	0.110
Lockdown Dummy Only	ARIMA(3,1,3)(0,0,0) [7]	11527.517	10035.263	11.632	7483.131	0.000
Double Seasonality (Period 14)	ARIMA(3,1,3)(2,0,1) [14]	6012.523	4819.490	5.436	6924.837	0.001

Descriptions and Rationale

Baseline: The original selected model, ARIMA(3,1,3)(2,0,1)[7], was identified through a comprehensive grid search for minimizing RMSE while maintaining low residual autocorrelation. It serves as the standard against which all refinements are evaluated.

Simplified Variant: This model reduces overall complexity by lowering both AR and MA orders. It was tested to determine if similar performance could be achieved with fewer parameters, potentially improving generalizability.

Reduce MA Order: The MA order was reduced from 3 to 2 while holding other terms constant. This tweak assessed whether the third MA term was contributing meaningfully to forecast accuracy.

Reduce AR Order: The AR order was reduced from 3 to 2. Like the MA tweak, this tests if fewer AR terms can still capture the underlying autocorrelation effectively.

Reduce Seasonal AR: The seasonal AR term was reduced from 2 to 1, evaluating whether the model was overfitting seasonal dynamics.

Reduce Seasonal MA: The seasonal MA term was removed to explore its impact on residual autocorrelation and forecast precision.

No Regressors: This version of the model omits all external regressors and simplifies the seasonal structure. It was included to assess how well an ARIMA model could perform based solely on internal time series dynamics, without the aid of dummy variables.

Weekend Dummy Only: This model excludes the lockdown dummy and retains only the weekend regressor. It tests whether modeling weekly patterns alone offers sufficient explanatory power.

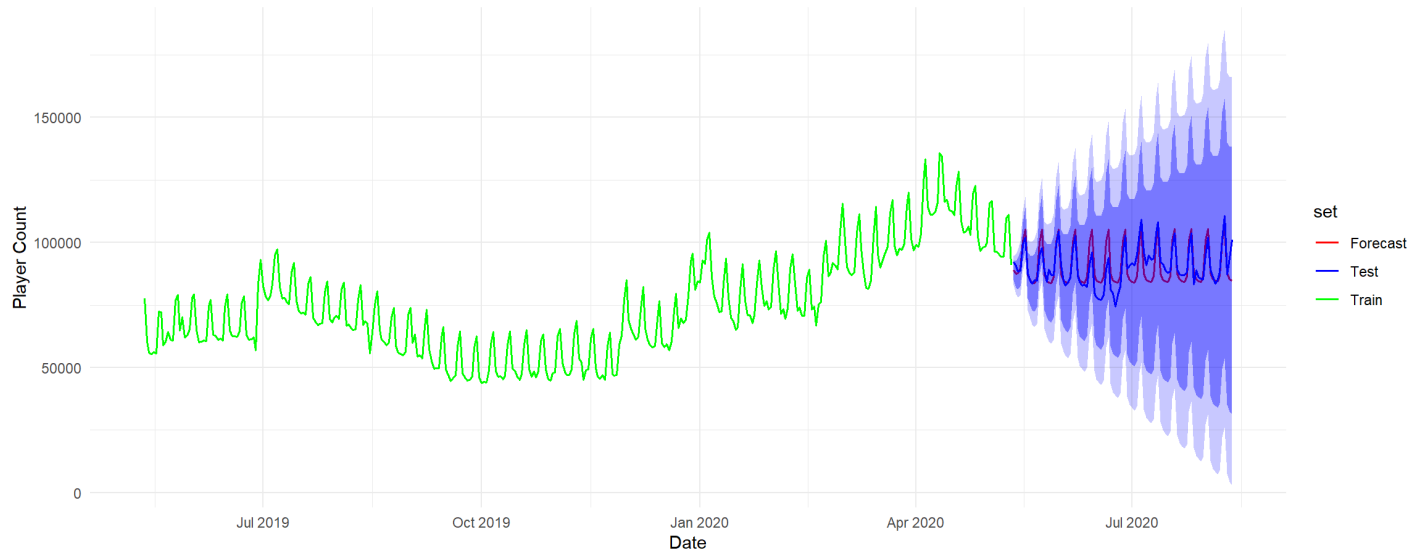
Lockdown Dummy Only: This model includes only the lockdown dummy as an external variable. It isolates the influence of the lockdown period on model performance while simplifying the seasonal terms.

Double Seasonality (Period 14): The seasonal period was changed from 7 to 14 to explore whether incorporating biweekly effects might improve forecast performance.

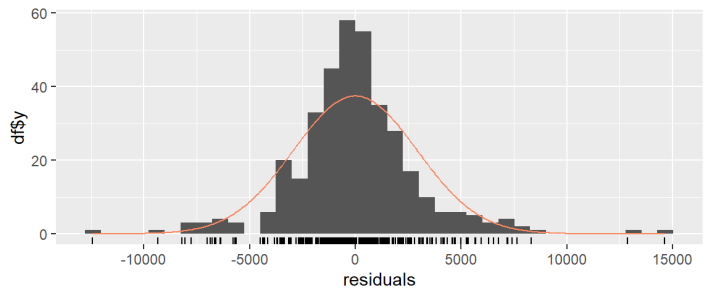
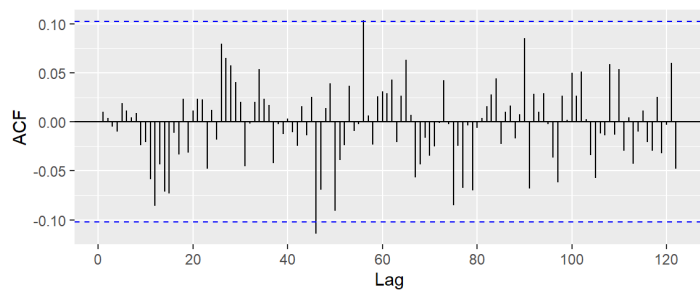
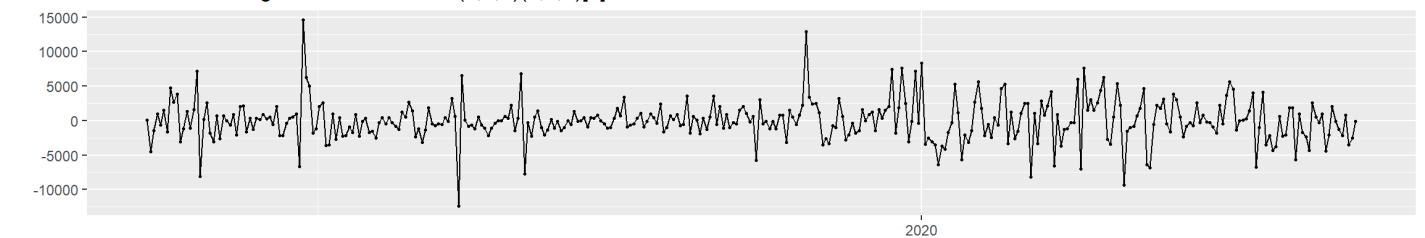
Final Model Selection: SARIMA(2,1,3)(2,0,1)[7]

After evaluating nine ARIMA model variations, ARIMA(2,1,3)(2,0,1)[7] was selected as the final forecasting model. While the baseline model ARIMA(3,1,3)(2,0,1)[7] achieved the lowest RMSE (4981.576), this alternate specification produced a nearly equivalent RMSE (5005.467) with two key advantages. It had a lower AICc value (6901.941 vs. 6904.127), suggesting a better trade-off between model fit and complexity, and a higher Ljung-Box p-value (0.114 vs. 0.062), indicating reduced residual autocorrelation. This version of the model uses one fewer autoregressive term, reducing complexity without a significant loss in forecast accuracy. Given these advantages, ARIMA(2,1,3)(2,0,1)[7] was selected for producing final forecasts and recommendation outputs.

Forecast from Final Model SARIMA(2,1,3)(2,0,1)[7] vs. Test Set

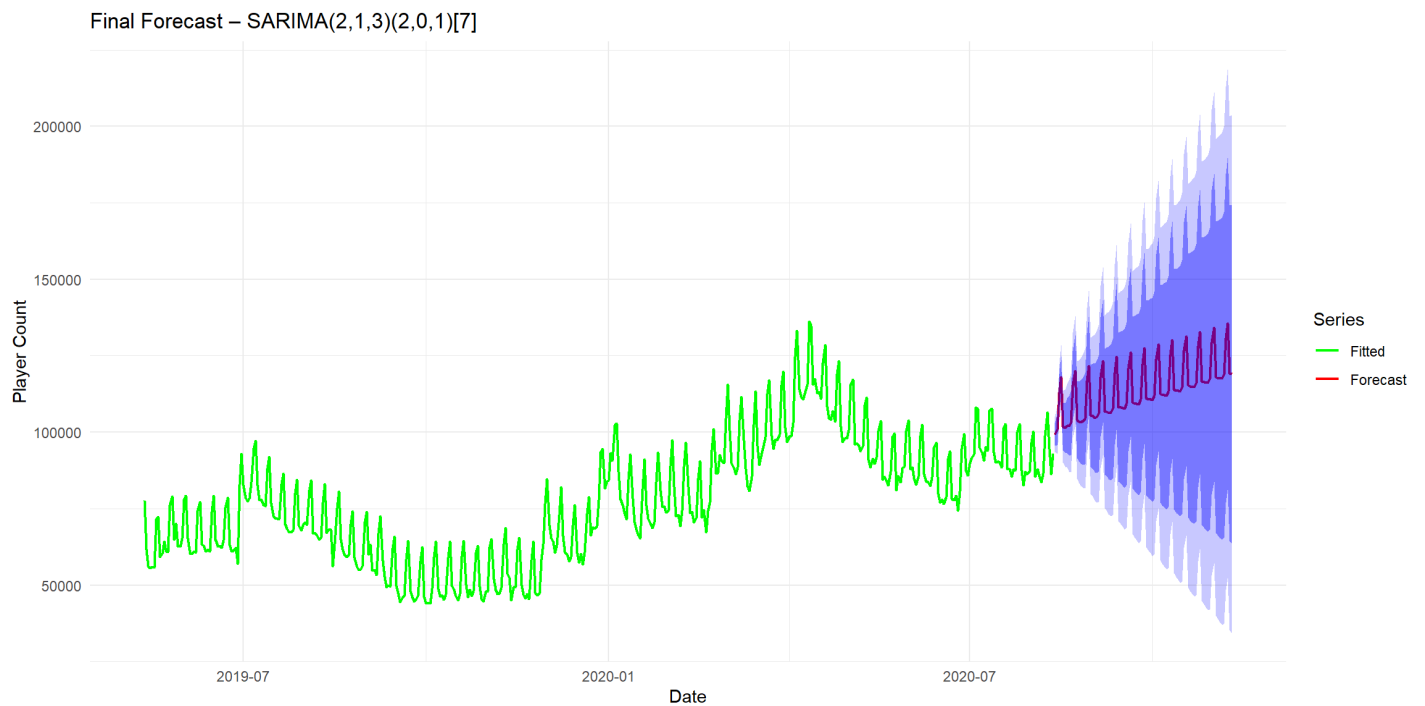


Residuals from Regression with ARIMA(2,1,3)(2,0,1)[7] errors



```
##
##  Ljung-Box test
##
## data:  Residuals from Regression with ARIMA(2,1,3)(2,0,1)[7] errors
## Q* = 49.12, df = 65, p-value = 0.9287
##
## Model df: 8.   Total lags used: 73
```

5) Implement Forecasts



Step 5 Commentary

The point forecasts from the final SARIMA model show a continued upward trend in player counts beyond the test period. While this trajectory aligns with elevated engagement during the COVID-19 lockdown period, it appears less reasonable when viewed in the context of the full series. Player counts were already beginning to decline by August 2020, suggesting the model may be overestimating future demand due to lingering effects of the temporary surge.

The forecast includes wide 80% and 95% prediction intervals that expand significantly over the 90-day horizon. This level of uncertainty seems appropriate given the volatility of the data and the limited post-lockdown observations available. It reflects the model's acknowledgment that future behavior could vary substantially—either continuing elevated engagement or reverting toward baseline levels.

One plausible scenario that would invalidate the forecasts is a post-pandemic normalization in player behavior. If the spike in engagement was temporary, driven by lockdown conditions, the continued upward trajectory in the forecasts would be unrealistic. Additionally, the emergence of new games or platform changes could shift player engagement dynamics in a way not captured by the historical data.

If future player counts fall well below forecasted values, the model could be improved by re-estimating on post-lockdown data only or incorporating a structural break. More granular covariates such as game releases and updates, holidays, or marketing events could also help improve predictive accuracy.